

Group Measurements

Combine Measures

```
pa_measures <- read.csv('../data/raw/Pennsylvania_State_Data/measures.csv') %>%
  mutate(format = tolower(format))
fl_measures <- read.csv('../data/raw/Florida_State_Data/measures.csv') %>%
  mutate(format = tolower(format))
or_measures <- read.csv('../data/raw/Oregon_State_Data/measures.csv') %>%
  mutate(format = tolower(format))

measures <- full_join(pa_measures, fl_measures, by = c("id", "name", "format")) %>%
  full_join(or_measures, fl_measures, by = c("id", "name", "format", "calculation"))

write_csv(measures, path = "measures_all.csv")

## Warning: The `path` argument of `write_csv()` is deprecated as of readr 1.4.0.
## i Please use the `file` argument instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

# all states combined, measures with filter_id == 1 (add defendants)
pa_m <- data_pa_comb %>% filter(filter_id == 1) %>%
  select(measure_id, measure_name) %>% distinct(measure_id, measure_name)
fl_m <- data_fl_comb %>% filter(filter_id == 1) %>%
  select(measure_id, measure_name) %>% distinct(measure_id, measure_name)
or_m <- data_or_comb %>% filter(filter_id == 1) %>%
  select(measure_id, measure_name) %>% distinct(measure_id, measure_name)

all_def_measures_all <- full_join(pa_m, fl_m, by = c("measure_id", "measure_name")) %>%
  full_join(or_m, fl_measures, by = c("measure_id", "measure_name"))

write_csv(all_def_measures_all, path = "all_def_measures_all.csv")
```

Distinct measure, filter pairs

```
data_pa_comb %>% distinct(measure_id, filter_id)
```

```
## # A tibble: 2,365 x 2
##   measure_id filter_id
##       <dbl>     <dbl>
## 1         7         1
## 2         7        10
## 3         7        11
## 4         7        12
```

```
## 5          7          13
## 6          7          14
## 7          7          15
## 8          7          16
## 9          7          20
## 10         7          21
## # i 2,355 more rows

# measures have multiple filters
```

Common Measures

```
chosen_filter <- 1 # "all defendants"

all_long <- bind_rows(data_pa_comb, data_fl_comb, data_or_comb) %>%
  filter(filter_id == chosen_filter, location_type == "county")

common_measures <- all_long %>%
  distinct(state, measure_id) %>%
  count(measure_id) %>%
  filter(n == 3) %>% # appears in all 3 states
  pull(measure_id)

# to remove N/As we find common measures - measures present in all three states
```

Pivot Wider

```
# We take all county data for all defedants measures

chosen_filter <- 1 # "all defendants"

pa_wide_m <- data_pa_comb %>%
  filter(filter_id == chosen_filter, location_type == "county", measure_id %in% common_measures) %>%
  select(filter_id, location_id, measure_name, value) %>%
  mutate(state = "PA") %>%
  pivot_wider(
    names_from = measure_name,
    values_from = value
  )

fl_wide_m <- data_fl_comb %>%
  filter(filter_id == chosen_filter, location_type == "county", measure_id %in% common_measures) %>%
  select(filter_id, location_id, measure_name, value) %>%
  mutate(state = "FL") %>%
  pivot_wider(
    names_from = measure_name,
    values_from = value
  )

or_wide_m <- data_or_comb %>%
  filter(filter_id == chosen_filter, location_type == "county", measure_id %in% common_measures) %>%
  select(filter_id, location_id, measure_name, value) %>%
```

```
mutate(state = "OR") %>%
pivot_wider(
  names_from = measure_name,
  values_from = value
)

all_wide_m <- bind_rows(pa_wide_m, fl_wide_m, or_wide_m)
```

Correlation Exploration Full (Ignore)

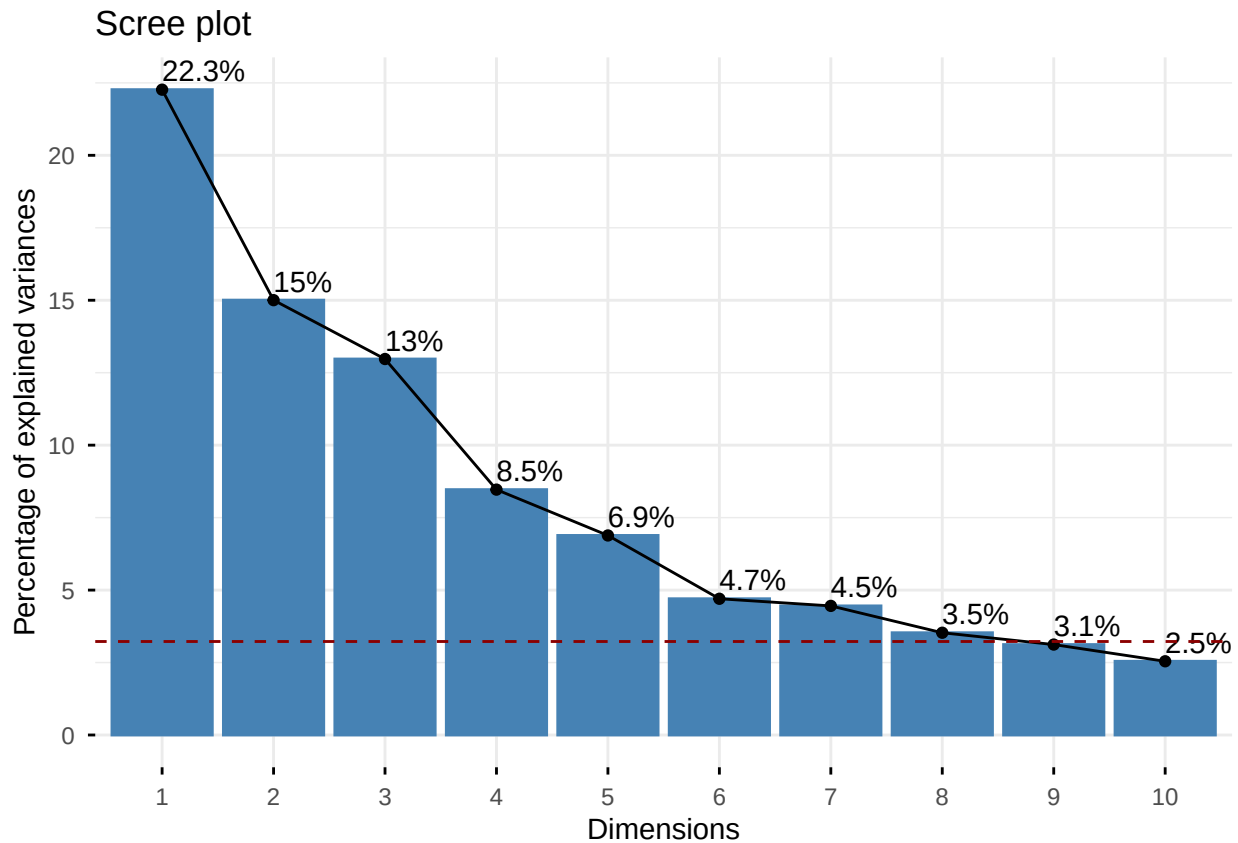
```
grouped_measures <- read_csv("all_def_measures_all_grouped.csv")

## Rows: 68 Columns: 4
## -- Column specification -----
## Delimiter: ","
## chr (2): measure_name, group_name
## dbl (2): measure_id, group_number
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
measures_quant.pre <- all_wide_m %>% drop_na()
measures_quant <- select(measures_quant.pre, -c(filter_id, location_id, state))
measures_quant.pca <- prcomp(measures_quant, center = TRUE, scale. = TRUE)
#summary(measures_quant.pca)

library(factoextra)

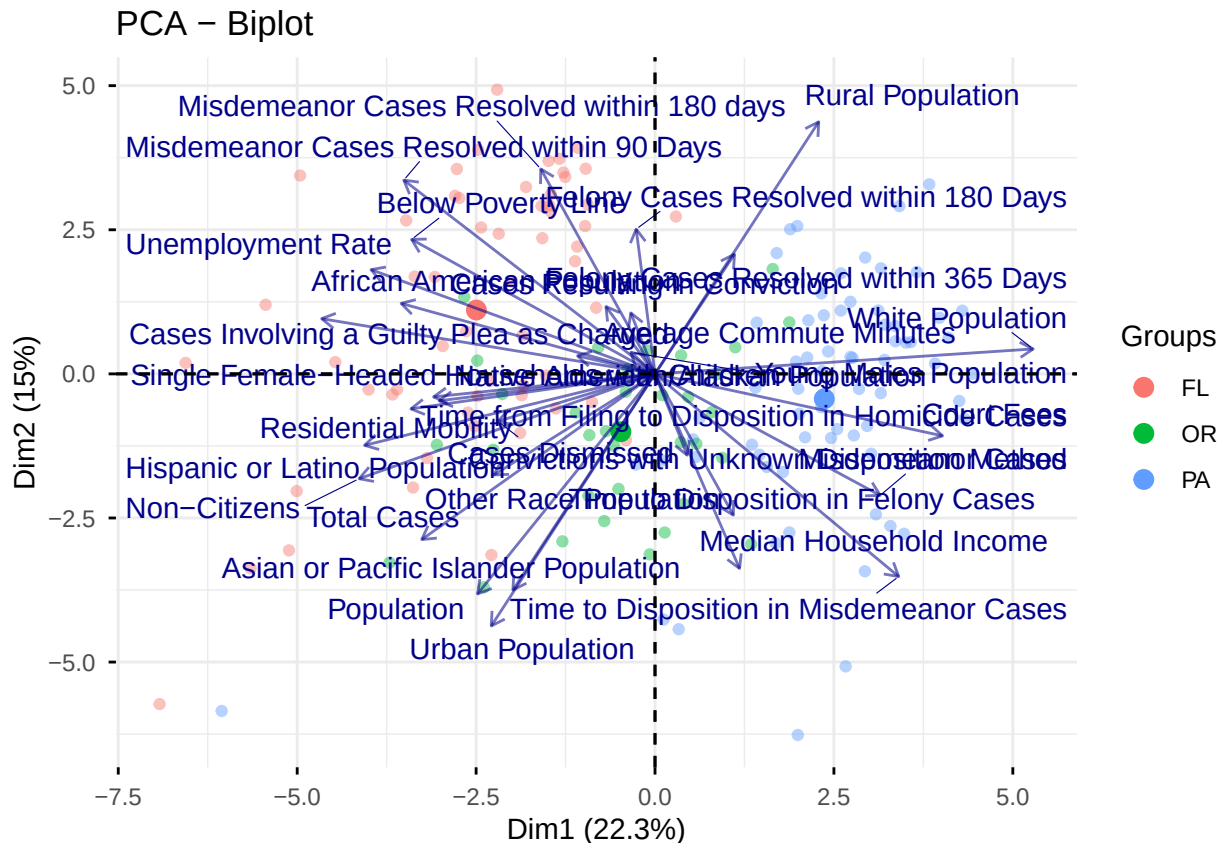
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
fviz_eig(measures_quant.pca, addlabels = TRUE) + # Add the labels
  # Have to multiply by 100 to get on percent scale
  geom_hline(yintercept = 100 * (1 / ncol(measures_quant)),
    linetype = "dashed", color = "darkred")

## Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :
## Ignoring empty aesthetic: `width`.
```



```
fviz_pca_biplot(measures_quant.pca,
  label = "var", # show variable names
  alpha.ind = 0.45, # transparency of points
  alpha.var = 0.55, # transparency of arrows
  col.var = "darkblue",
  repel = TRUE,
  # Set the color of the points to decades variable:
  habillage = measures_quant.pre$state, pointshape = 19
)
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## i The deprecated feature was likely used in the ggpubr package.
## Please report the issue at <https://github.com/kassambara/ggpubr/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



The scree plot indicates that no single principal component dominates the variance structure. The first component explains 22.3% of total variation, and the first three components together account for approximately 50%. This suggests that the data contain multiple underlying dimensions rather than a single dominant factor.

Groupings

1 = Pretrial Release & Bail Practices 2 = Case Processing Speed & Court Efficiency 3 = Adjudication Patterns 4 = Sentencing Severity & Incarceration Decisions 5 = Financial Penalties & Legal Debt 6 = Post-Sentencing Outcomes 7 = Court & Law Enforcement Capacity 8 = Demographic Structure 9 = Socioeconomic Conditions

PCA by Category

```
group_list <- split(grouped_measures$measure_name, grouped_measures$group_name)

# get measures for each group
measures_by_group <- lapply(group_list, function(vars) {
  vars_in_data <- vars[vars %in% colnames(measures_quant.pre)]
  measures_quant.pre %>% select(state, any_of(vars_in_data))
})

# ex. extract measure for a given group -- measures_by_group[["Socioeconomic Conditions"]]

plot_pca_biplot <- function(m_quant, group) {
  # create pcs biplot for the given data m_quant, representing group
}
```

```

measures_quant <- select(m_quant, -c(state))
measures_quant.pca <- prcomp(measures_quant, center = TRUE, scale. = TRUE)
#summary(measures_quant.pca)

fviz_eig(measures_quant.pca, addlabels = TRUE) + # Add the labels
# Have to multiply by 100 to get on percent scale
  geom_hline(yintercept = 100 * (1 / ncol(measures_quant)),
    linetype = "dashed", color = "darkred")

return (list(pca_plot =
  fviz_pca_biplot(measures_quant.pca,
    label = "var", # show variable names
    alpha.ind = 0.45, # transparency of points
    alpha.var = 0.55, # transparency of arrows
    col.var = "darkblue",
    repel = TRUE,
    # Set the color of the points to decades variable:
    habillage = m_quant$state, pointshape = 19
  ) +
  labs(title = paste("PCA - Biplot:", group)),
  rot = measures_quant.pca$rotation,
  m_quant = measures_quant
))
}

# the commented out plots have no data (or only 1 measure) -> cannot make plot

#plot_pca_biplot(measures_by_group[["Pretrial Release & Bail Practices"]], "Pretrial Release & Bail Practices")
pca2<- plot_pca_biplot(measures_by_group[["Case Processing Speed & Court Efficiency"]], "Case Processing Speed & Court Efficiency")

## Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :
## Ignoring empty aesthetic: `width`.
pca3 <- plot_pca_biplot(measures_by_group[["Adjudication Patterns"]], "Adjudication Patterns")

## Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :
## Ignoring empty aesthetic: `width`.
#plot_pca_biplot(measures_by_group[["Sentencing Severity & Incarceration Decisions"]], "Sentencing Severity & Incarceration Decisions")

# contains one measure (Court Fees)
#plot_pca_biplot(measures_by_group[["Financial Penalties & Legal Debt"]], "Financial Penalties & Legal Debt")

#plot_pca_biplot(measures_by_group[["Post-Sentencing Outcomes"]], "Post-Sentencing Outcomes")
pca7 <- plot_pca_biplot(measures_by_group[["Court & Law Enforcement Capacity"]], "Court & Law Enforcement Capacity")

## Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :
## Ignoring empty aesthetic: `width`.
pca8 <- plot_pca_biplot(measures_by_group[["Demographic Structure"]], "Demographic Structure")

## Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :
## Ignoring empty aesthetic: `width`.
pca9 <- plot_pca_biplot(measures_by_group[["Socioeconomic Conditions"]], "Socioeconomic Conditions")

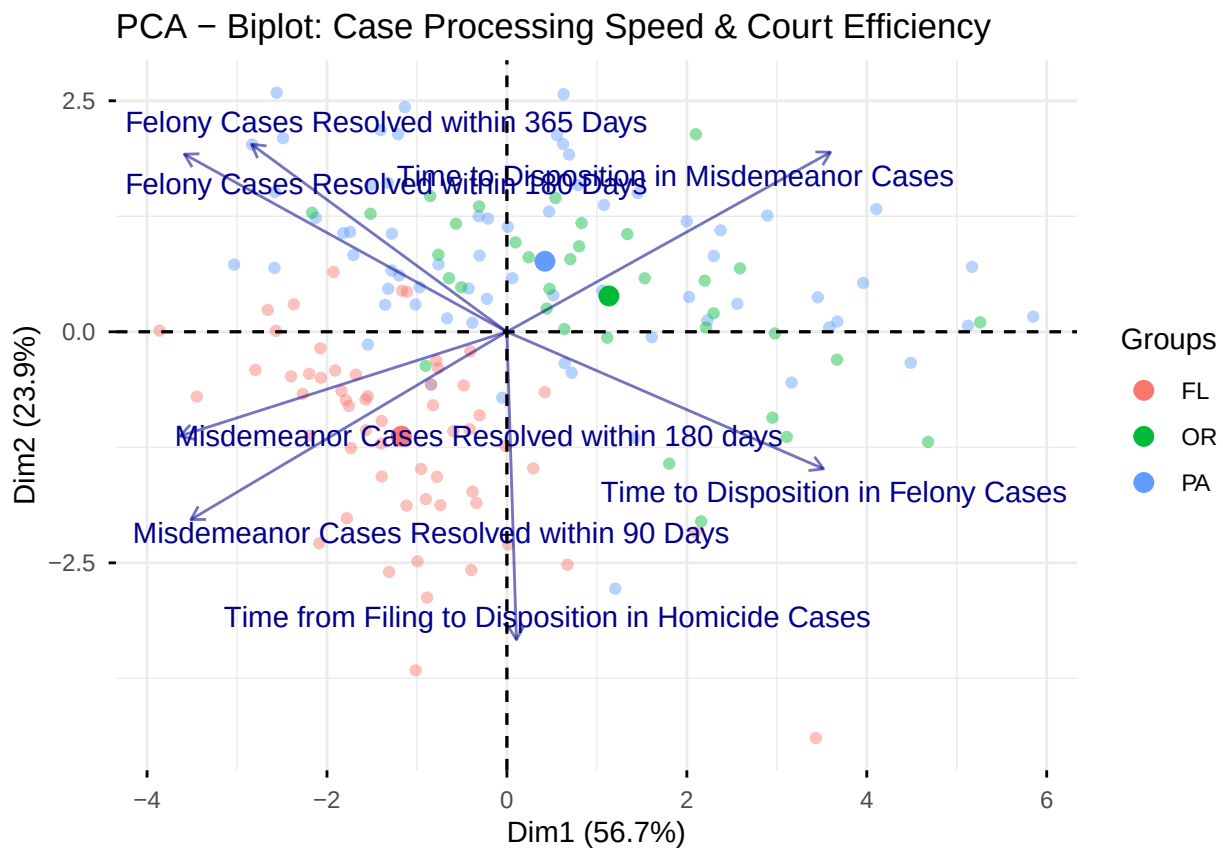
## Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :

```

```
## Ignoring empty aesthetic: `width`.
measure_groups_df <- read_csv("measure_groups.csv")

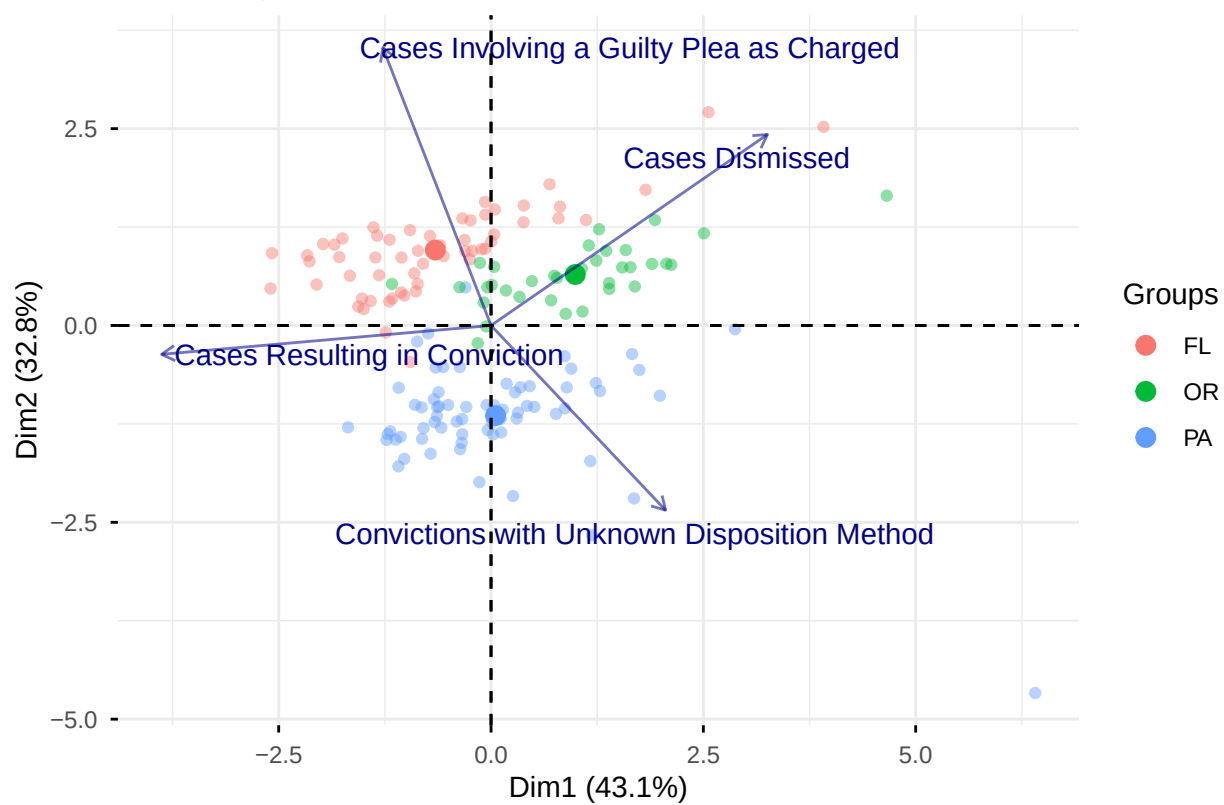
## Rows: 9 Columns: 2
## -- Column specification -----
## Delimiter: ","
## chr (1): group_name
## dbl (1): group_id
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
#lapply(measure_groups_df$group_name, function(g) {
#  plot_pca_biplot(measures_by_group[[g]], g)
#})
```

```
pca2$pca_plot
```



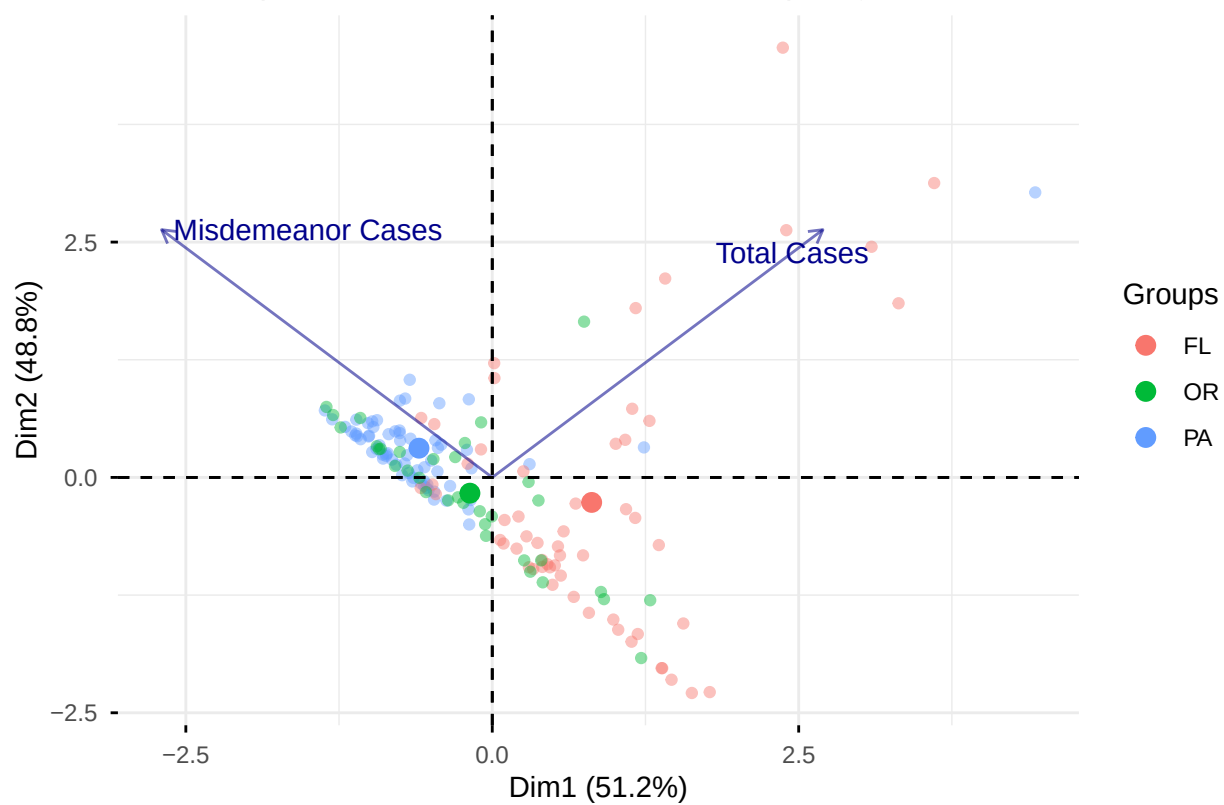
```
pca3$pca_plot
```

PCA – Biplot: Adjudication Patterns



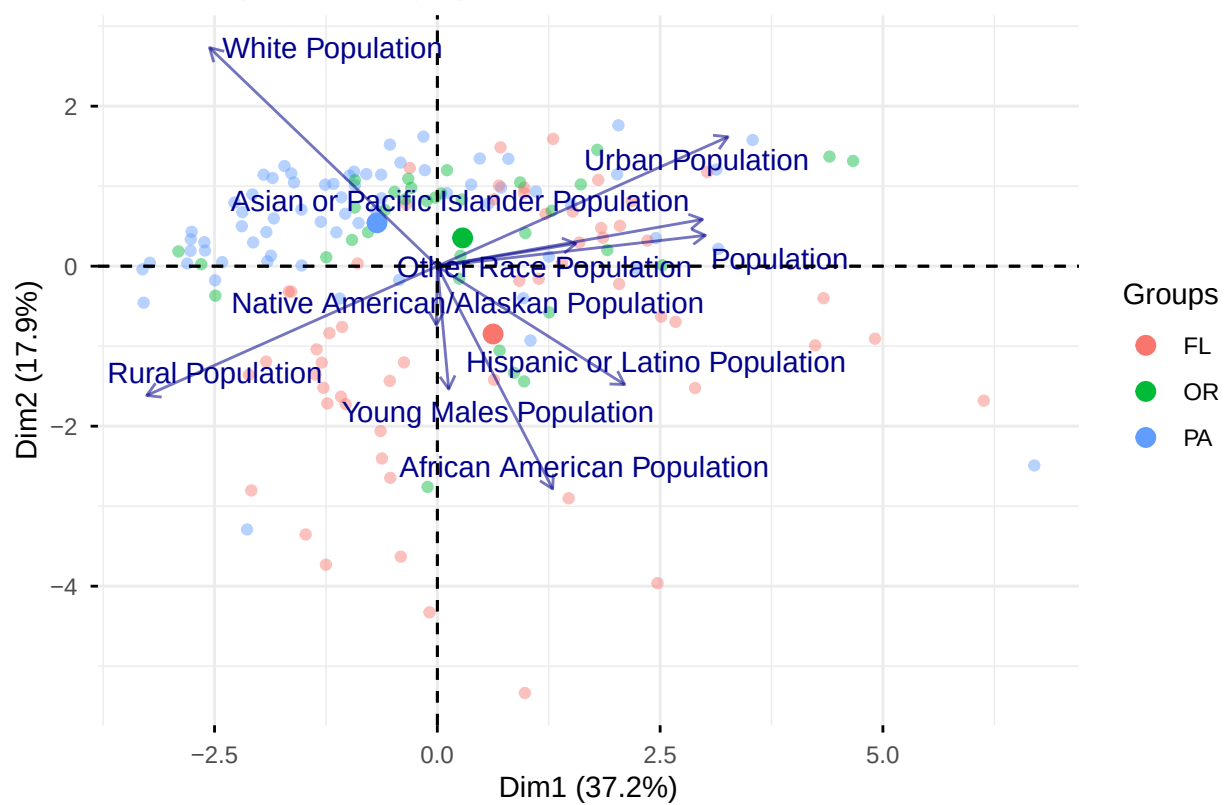
pca7\$pca_plot

PCA – Biplot: Court & Law Enforcement Capacity

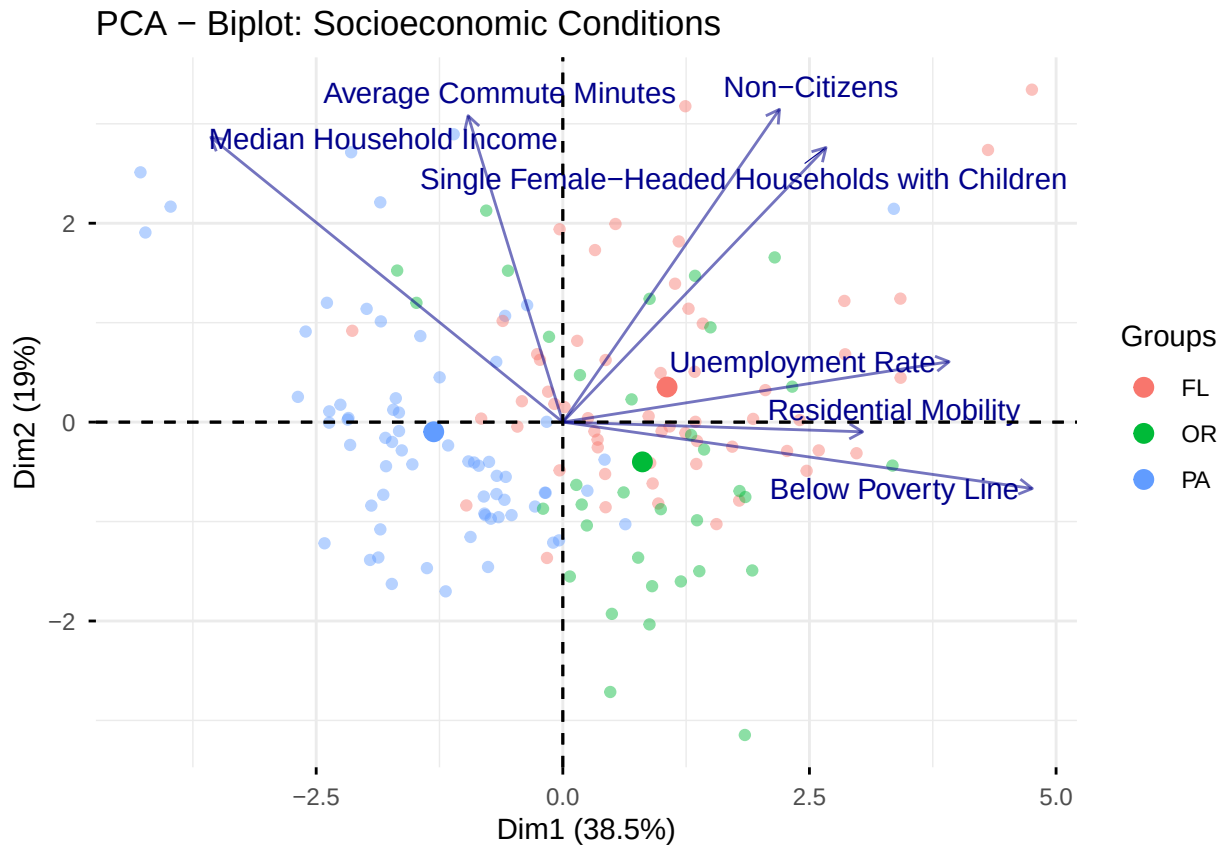


```
pca8$pca_plot
```

PCA – Biplot: Demographic Structure



pca9\$pca_plot



2. Case Processing Speed & Court Efficiency

This plot has Dim1 = 56.7%, so PC1 is doing a lot of the work. The main split appears to be between counties with longer case-processing times on the right and counties with faster resolution / higher percent resolved within benchmark days on the left. In particular, “Time to Disposition in Felony Cases” and “Time to Disposition in Misdemeanor Cases” point strongly to the right, while measures like “Felony Cases Resolved within 180 Days,” “Felony Cases Resolved within 365 Days,” and the misdemeanor resolved-within-X-days measures point left. So PC1 seems to capture an overall slow vs. fast court processing dimension. PC2 adds a secondary distinction, separating some felony-resolution measures upward from homicide/misdemeanor timing measures downward, but the dominant story is still overall case-processing speed.

3. Adjudication Patterns

This one has Dim1 = 43.1% and Dim2 = 32.8%, so there is no single dominant dimension. The category is more clearly two-dimensional. Along PC1, the main contrast seems to be between dismissals / unknown disposition method on the right and cases resulting in conviction on the left. Along PC2, “Cases Involving a Guilty Plea as Charged” points strongly upward while “Convictions with Unknown Disposition Method” points strongly downward. So this category is not just one clean continuum; it reflects multiple adjudication patterns.

7. Court & Law Enforcement Capacity

This is the cleanest one. There are only two measures shown: Total Cases and Misdemeanor Cases. PC1 explains 51.2% and PC2 explains 48.8%, which makes sense with essentially two variables. The arrows point in fairly different directions, so the two measures are capturing somewhat distinct variation, not one single shared axis. The county clustering suggests state differences in case-volume structure, with Florida more spread toward the “Total Cases” direction and PA/OR more concentrated near the center-left.

8. Demographic Structure

This plot is clearly more complex. Dim1 = 37.2% and Dim2 = 17.9%, so PC1 matters, but there is still a lot of remaining variation outside these first two dimensions. The horizontal axis seems to separate counties with higher urban / total population / Hispanic or Latino population / African American population toward the right from counties with higher white population / rural population toward the left. PC2 seems to further distinguish white population upward from some other demographic measures downward. So the first component looks like a rough urban-diverse vs. white-rural contrast.

9. Socioeconomic Conditions This is a fairly interpretable plot. Dim1 = 38.5% and Dim2 = 19%. PC1 seems to separate counties with higher poverty, unemployment, residential mobility, single-female-headed households with children, non-citizens on the right from counties with higher median household income on the left. That looks like a pretty classic socioeconomic disadvantage gradient. PC2 adds something like a household/commuting/family-structure dimension, but the main story is the left-right gradient of advantage vs. disadvantage.

we look at rotation for the dominating PC (usually PC1) to find the "top" measure of each category

```
sort(abs(pca2$rot[, "PC1"]), decreasing = TRUE)
```

```
##      Misdemeanor Cases Resolved within 180 days
##                                0.42883003
##      Time to Disposition in Misdemeanor Cases
##                                0.42449948
##      Felony Cases Resolved within 180 Days
##                                0.42319132
##      Time to Disposition in Felony Cases
##                                0.41533046
##      Misdemeanor Cases Resolved within 90 Days
##                                0.41455458
##      Felony Cases Resolved within 365 Days
##                                0.33511701
## Time from Filing to Disposition in Homicide Cases
##                                0.01248243
```

Felony Cases Resolved within 180 Days

```
sort(abs(pca3$rot[, "PC1"]), decreasing = TRUE)
```

```
##      Cases Resulting in Conviction
##                                0.6912318
##      Cases Dismissed
##                                0.5800506
## Convictions with Unknown Disposition Method
##                                0.3663386
##      Cases Involving a Guilty Plea as Charged
##                                0.2270152
```

Cases Resulting in Conviction

```
pca7$rot
```

```
##      PC1      PC2
## Total Cases    0.7071068 0.7071068
## Misdemeanor Cases -0.7071068 0.7071068
```

Total Cases

```
sort(abs(pca8$rot[, "PC1"]), decreasing = TRUE)
```

```
##           Urban Population           Rural Population
##           0.442836472           0.442836472
##           Population Asian or Pacific Islander Population
##           0.408505987           0.404056920
##           White Population           Hispanic or Latino Population
##           0.346995311           0.285388711
##           Other Race Population           African American Population
##           0.211188663           0.175768590
##           Young Males Population           Native American/Alaskan Population
##           0.017382247           0.001651892
```

```
# Population
```

```
sort(abs(pca9$rot[, "PC1"]), decreasing = TRUE)
```

```
##           Below Poverty Line
##           0.5577528
##           Unemployment Rate
##           0.4589035
##           Median Household Income
##           0.4181122
##           Residential Mobility
##           0.3561111
## Single Female-Headed Households with Children
##           0.3126886
##           Non-Citizens
##           0.2574034
##           Average Commute Minutes
##           0.1124981
```

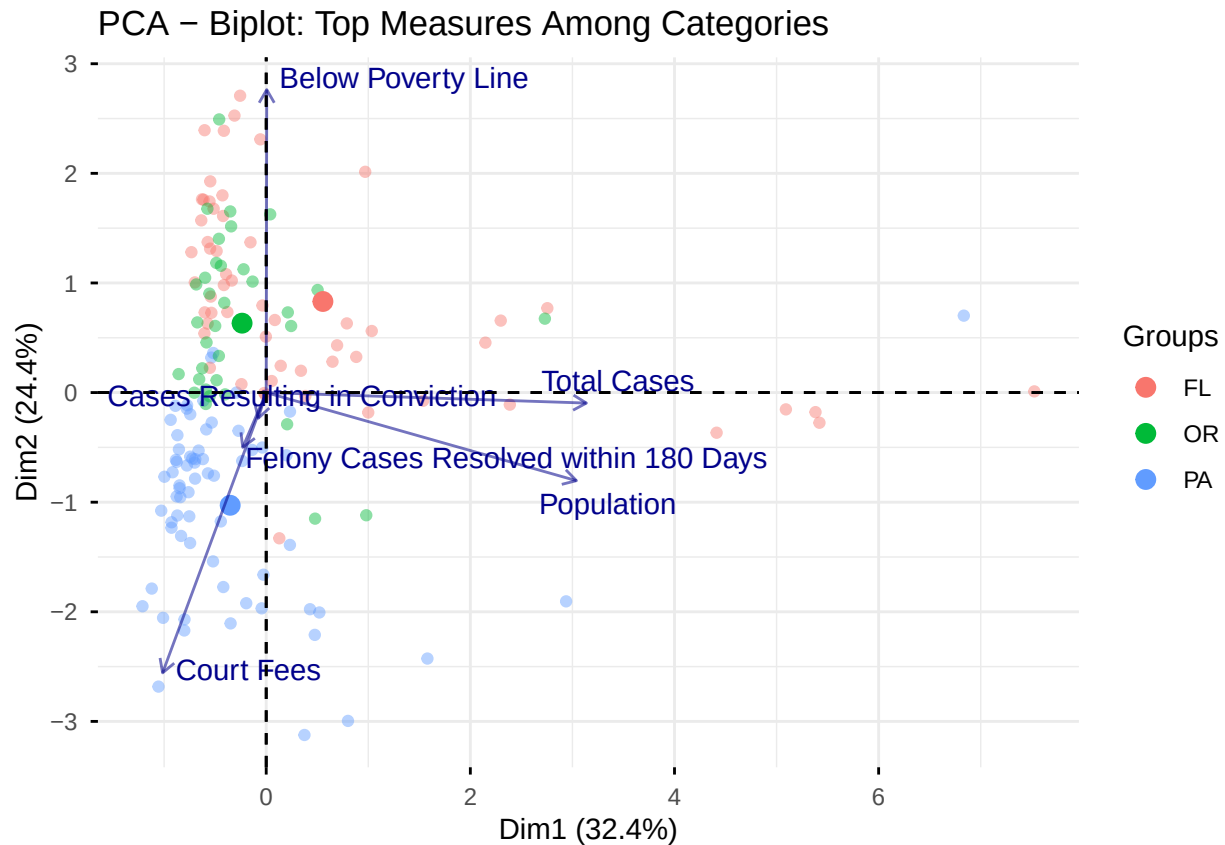
```
# Below Poverty Line
```

```
top_measures = c("Felony Cases Resolved within 180 Days",
  "Cases Resulting in Conviction",
  "Total Cases",
  "Population",
  "Below Poverty Line",
  "Court Fees")
```

```
measures_quant.pre.top <- measures_quant.pre %>% select(state, any_of(top_measures))
pcaTop <- plot_pca_biplot(measures_quant.pre.top, "Top Measures Among Categories")
```

```
## Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :
## Ignoring empty aesthetic: `width`.
```

```
pcaTop$pca_plot
```



Heat Map

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
cor_matrix <- cor(pca2$m_quant, use = "complete.obs")
corrplot(cor_matrix,
  method = "color",
  type = "upper",
  tl.col = "black",
  tl.srt = 45)
```

```
## Warning in corrplot(cor_matrix, method = "color", type = "upper", tl.col =
## "black", : Not been able to calculate text margin, please try again with a
## clean new empty window using {plot.new(); dev.off()}} or reduce tl.cex
```

Misdemeanor Cases Resolved within 180 Days
 Felony Cases Resolved within 180 Days
 Time from Filing to Disposition in Homicide Cases

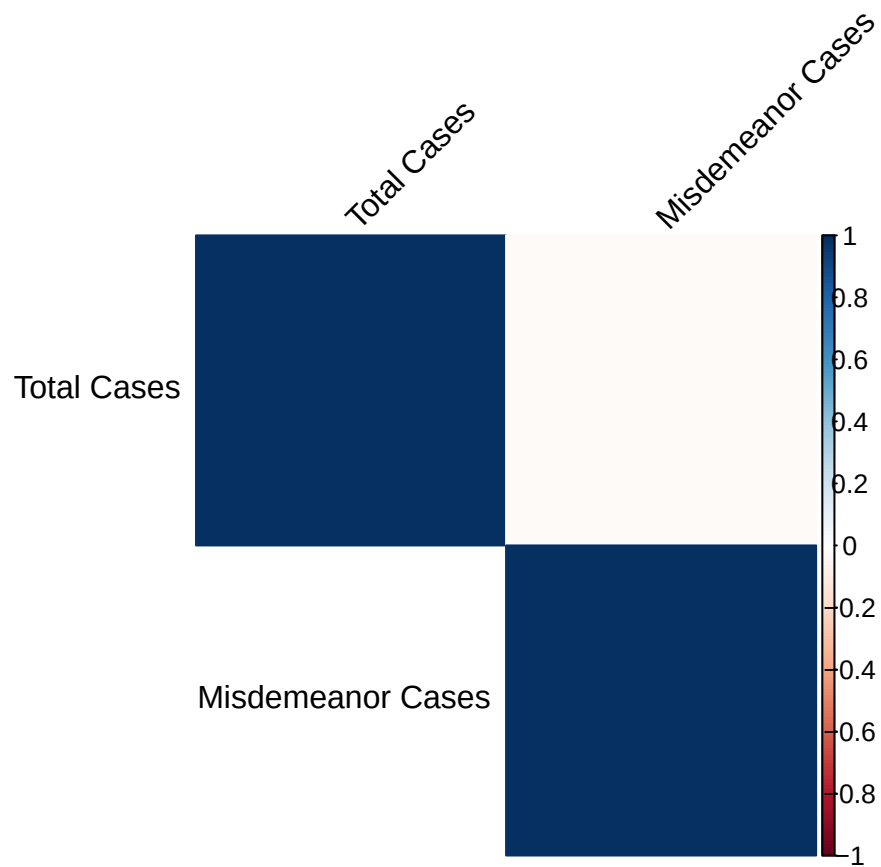
```
cor_matrix <- cor(pca3$m_quant, use = "complete.obs")
corrplot(cor_matrix,
  method = "color",
  type = "upper",
  tl.col = "black",
  tl.srt = 45)
```

```
## Warning in corrplot(cor_matrix, method = "color", type = "upper", tl.col =
## "black", : Not been able to calculate text margin, please try again with a
## clean new empty window using {plot.new(); dev.off()} or reduce tl.cex
```

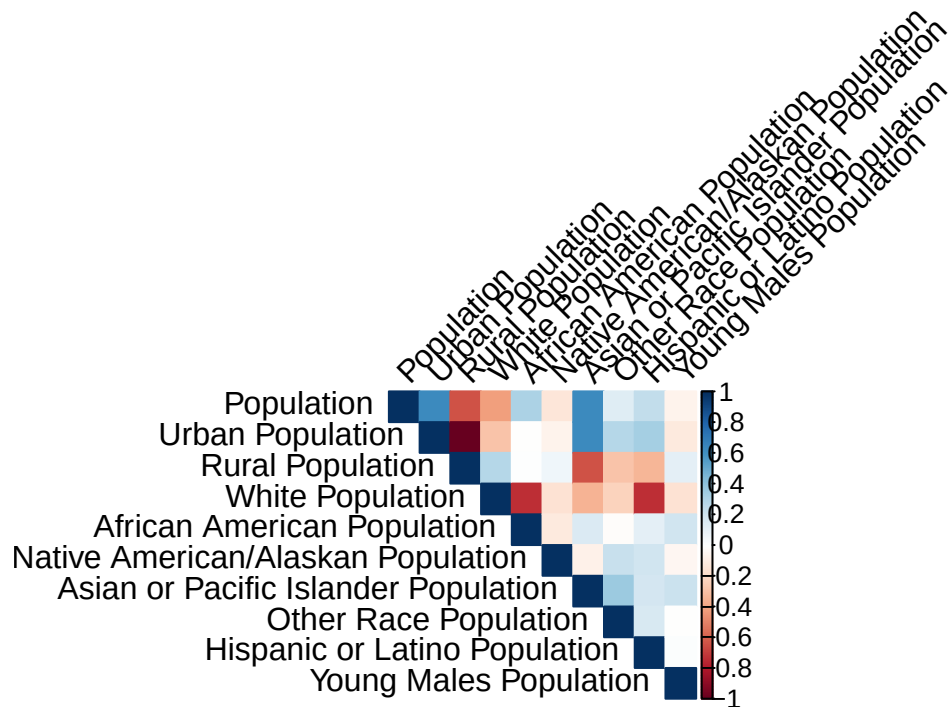
Cases Dismissed
 Convictions with Unknown Disposition Method
 Cases Involving a Guilty Plea as Charged
 Cases Resulting in Conviction

```
cor_matrix <- cor(pca7$m_quant, use = "complete.obs")
corrplot(cor_matrix,
  method = "color",
  type = "upper",
  tl.col = "black",
```

```
tl.srt = 45)
```



```
cor_matrix <- cor(pca8$m_quant, use = "complete.obs")  
corrplot(cor_matrix,  
  method = "color",  
  type = "upper",  
  tl.col = "black",  
  tl.srt = 45)
```



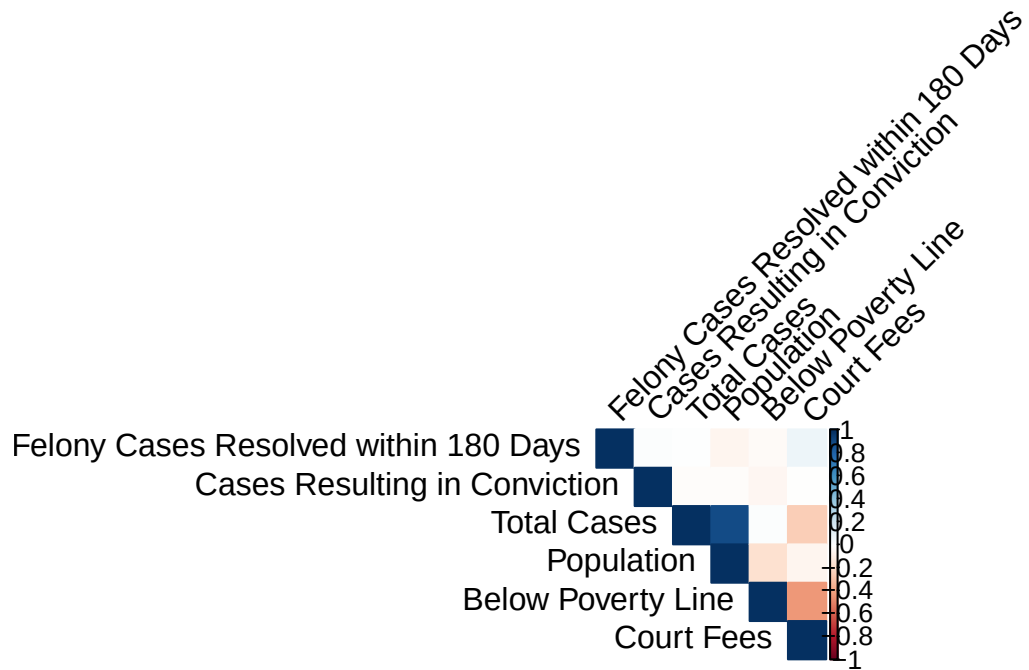
```
cor_matrix <- cor(pca9$m_quant, use = "complete.obs")
corrplot(cor_matrix,
  method = "color",
  type = "upper",
  tl.col = "black",
  tl.srt = 45)
```

```
## Warning in corrplot(cor_matrix, method = "color", type = "upper", tl.col =
## "black", : Not been able to calculate text margin, please try again with a
## clean new empty window using {plot.new(); dev.off()} or reduce tl.cex
```

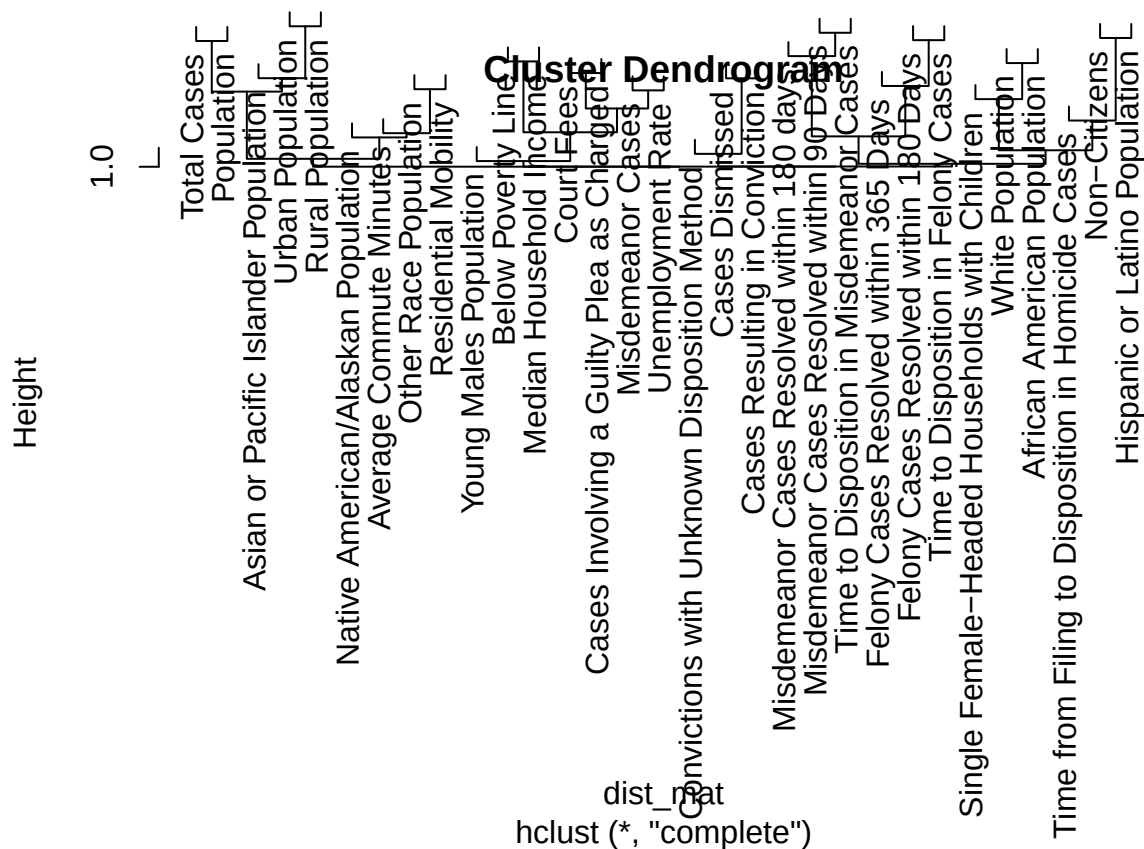
Single Female-Headed Households with Children
 Median Household Income
 Average Residential Mobility

```
library(corrplot)
cor_matrix <- cor(pcaTop$m_quant, use = "complete.obs")
```

```
corrplot(cor_matrix,
  method = "color",
  type = "upper",
  tl.col = "black",
  tl.srt = 45)
```



```
dist_mat <- as.dist(1 - abs(cor(measures_quant)))
hc <- hclust(dist_mat)
plot(hc)
```



```
library(factoextra)

par(mar = c(15, 4, 4, 2)) # bigger bottom margin

make_dend <- function(m_quant) {
  measures_scaled_quant <- m_quant %>%
    scale(center = FALSE, scale = apply(., 2, sd, na.rm = TRUE))
  measures_dist <- dist(t(measures_scaled_quant))

  hc_measures_complete <- hclust(measures_dist, method = "complete")
  hc_measures_complete_dend <- as.dendrogram(hc_measures_complete)
  plot(hc_measures_complete_dend, main = "Complete Linkage")

  hc_measures_single <- hclust(measures_dist, method = "single")
  hc_measures_single_dend <- as.dendrogram(hc_measures_single)
  plot(hc_measures_single_dend, main = "Single Linkage")
}

make_dend(pca2$m_quant)
```

Complete Linkage

